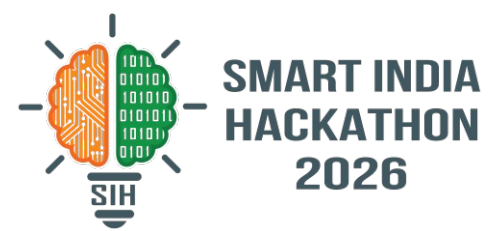


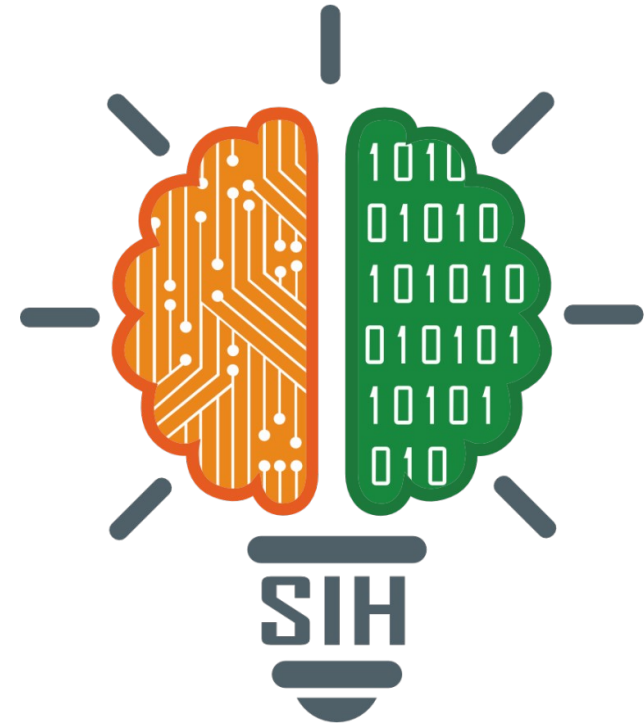


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26153
- **Problem Statement Title** – AI based Network Attack Forecasting from Network Traffic Data
- **Theme** – Blockchain and Cybersecurity
- **PS Category** – Software
- **Team ID** – 149775
- **Team Name (Registered on portal)** – Resurrección





NetSentinel Foresight : A World Model that Forecasts Network Attacks



Enterprise & Critical Information Infrastructure | Learns $P(S_{t+1} | S_t)$ | K-step attack forecasting | MITRE ATT&CK stages | SHAP + attention explanations | Runs fully offline

IDEA / SOLUTION

Intrusion detectors tell you a breach has happened. **NetSentinel Foresight tells you where the attacker is going next, while the attacker is still scanning and there is still time to stop them.**



Learns how the network changes over time

A world model learns $P(S_{t+1} | S_t)$: how a graph of hosts, flow and packet features changes each minute. It does not just label single flows.



Plays the future forward

Simulates 64 possible futures up to K windows ahead. The share that reach an infiltration state is the infiltration probability, shown with a confidence band.



Names the next attack stage and target

Each forecast maps to MITRE ATT&CK: Reconnaissance → Initial Access → Lateral Movement → C2 → Exfiltration, and names the host likely hit next.



Shows its reasoning

Attention weights and SHAP values list the flags, ports and flow statistics behind every forecast. No unexplained scores.



“What if we block it?”

The defender applies an action (block port 445, isolate a host) inside the model and sees how the risk changes, before touching the real network.



Built on a pipeline that already works

Reuses our tested PCAP→feature extractor, six detectors and signed, tamper-evident alert ledger. Runs offline on an ordinary CPU.

PROBLEM RESOLUTION

- **Classifiers see flows, not campaigns:** a benign/malicious label per flow throws away order and timing. A slow scan, then a brute-force, then lateral SMB traffic is one story, not three unrelated alerts.
- **Alerts come after the damage:** detection fires once a stage has already happened. Defenders need the warning while the attacker is still in reconnaissance.
- **Scores nobody can act on:** a probability with no “why” and no “what next” gets ignored by SOC analysts and cannot be audited in CII.

INNOVATION & UNIQUENESS



Attack “weather forecast”

Many sampled futures give a risk band and time-to-compromise, not one label.



Flow + packet, on a graph

TTL, window size and port order catch slow scans; the graph predicts which host is next.



Auditable forecasts

Each forecast is signed into a Merkle ledger and later scored against what actually happened.



Knows the attacker’s playbook

ATT&CK/CAPEC stage prior + CVE exposure guide the dynamics, so it copes with attacks it never saw.



What-if defence simulator

Test a block or isolate action inside the model before using it.



Cheap enough for CII

A small Sentry scores every host; the rollouts run only for the few it escalates. CPU only.

Built: PCAP/CSV feature pipeline
tested on real traffic (Tier 1)

Next: world model trained
CIC-IDS2018 + CTU-13 + UNSW-NB15

Next: K-step forecaster + explainer
offline Streamlit demo

Next: multi-stage validation
DARPA 2000 + LANL red team, CII pilot



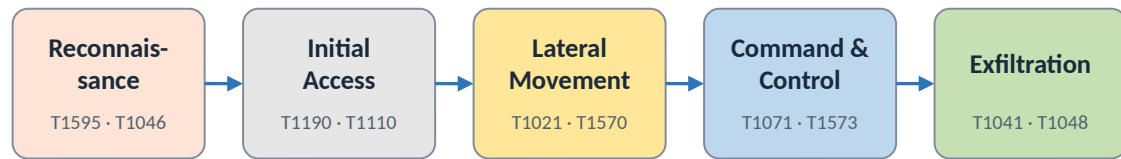
TECHNICAL APPROACH

WORLD MODEL ARCHITECTURE

NetSentinel Foresight learns **how network state changes**. An encoder turns each time window into a state, a dynamics model learns $P(S_{t+1} | S_t)$, and a rollout engine simulates K steps ahead.

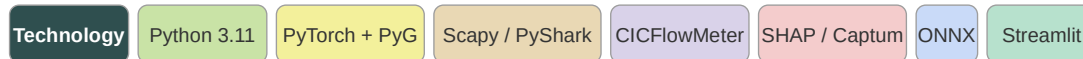
- **State S_t (every 60 s window):** a graph with hosts as nodes and flows as edges. Flow-level features: TCP flag counts (SYN/ACK/FIN/RST/PSH/URG), bytes, packets, duration, IAT mean/var/max, bidirectional ratio. Packet-level features: TTL mean and variance, TCP window size, fragment flags, payload-size histogram, sequential vs random port order, retransmissions.
- **Encoder:** E-GraphSAGE edge aggregation (reused from our Tier-2 Inspector) → a latent vector for each host and for the whole network.
- **Dynamics (the world model):** recurrent state-space model (GRU memory + stochastic latent) with a causal Temporal Transformer over the last 30 windows. It learns to predict the next state's features (self-supervised) and the next ATT&CK stage from the datasets' attack timelines (supervised dynamics learning).
- **Forward simulation:** sample 64 latent paths $K = 1 \dots 10$ windows ahead → $P(\text{infiltration within } K)$, predicted stage and next target host.
- **Explanations:** attention over time and neighbours + SHAP over feature groups → e.g. "top drivers: SYN-only ratio ↑, dst-port entropy ↑, TTL variance ↑".

FORECAST ATTACK STAGES (MITRE ATT&CK)

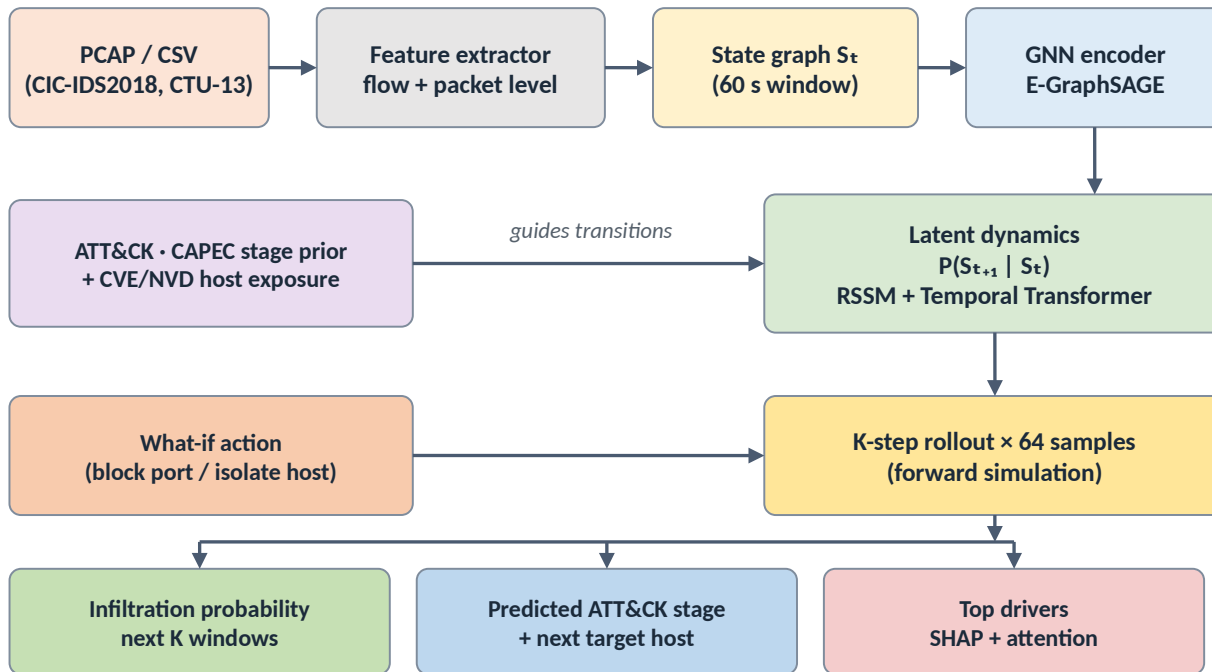


Dataset labels (e.g. CIC-IDS2018 "Infiltration", "Brute Force", "Bot"; DARPA 2000 LLDOS phases 1-5) are mapped to these stages. The benign → recon → access transitions are the ones the model must learn to see coming.

NETSENTINEL TECH STACK



WORLD-MODEL PIPELINE



Every forecast is signed into our Merkle ledger, so it can be checked later against what actually happened.

Supervised dynamics learning: ground-truth transitions come from each dataset's attack timeline. The loss weights transitions into attack stages, so rare stages are learned. **Output per window:** $P(\text{infiltration in the next } K \text{ windows})$, predicted stage, top contributing flags/ports/flow stats, and the next target host. **Not a static classifier:** we show it by shuffling time. If accuracy does not drop, it has not learned dynamics.



FEASIBILITY & VIABILITY

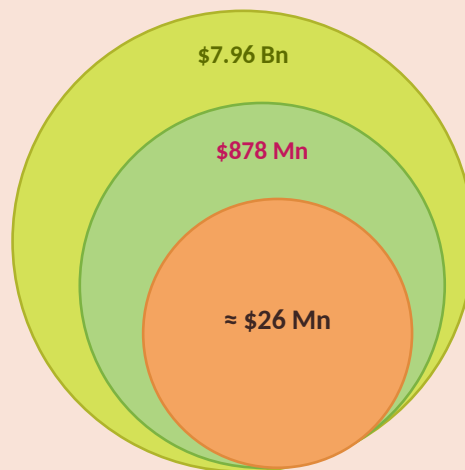
POTENTIAL CHALLENGES AND RISKS:

Attack windows are rare: attack time is a small fraction of each capture, and stages like exfiltration are rarer still.

Order may carry no signal: in our own earlier test, shuffling time changed AUC by 0.000 on synthetic data. A world model has to prove it beats a static one.

Dataset shift: a model trained on a lab network may not carry over to a real enterprise or CII network.

Market Opportunity Funnel



TAM \$7.96 Bn

Global Intrusion Detection & Prevention market, 2026
(Precedence Research)

SAM up to \$878 Mn

India OT security market, 2026
(MarketsandMarkets)

SOM ≈ \$26 Mn

We aim for 3% of the available market: SOCs that want early warning, not only alerts.

STRATEGIES TO OVERCOME THESE CHALLENGES:

- **Weight the rare stages:** focal loss, oversampling of attack transitions, training on several datasets (CIC-IDS2018, CTU-13, UNSW-NB15, CICIOT2023).
- **Prove the time effect:** every result is shown next to logistic regression on the same features and next to our model with time shuffled. If there is no gap, we say so.
- **No-leak, unseen-attack tests:** split by time, never by random row. Hold out whole attack families and whole datasets. Fine-tune per site on ~10 days of its own benign traffic.

Component	Trained / tested on	What we will report
Feature pipeline · Scapy + CICFlowMeter	CIC-IDS2018 CSV + PCAP, CTU-13	Timestamped, normalised flow + packet matrix; flows/s
World model · RSSM + Transformer + GNN	CIC-IDS2018, CTU-13, UNSW-NB15	Next-state error; stage F1 at $t+1 \dots t+K$
Forecast engine · K-step rollout	DARPA 2000 LLDOS, LANL red team	Lead time before compromise; AUC@K; Brier score
Baseline · Logistic regression	Same features, same time split	F1, precision, recall, FPR, side by side
Generalisation	Leave-one-attack-out; CICIOT2023	Recall on attack types never seen in training
Explainer · SHAP + attention	All of the above	Deletion test: remove top-5 drivers → risk must fall

PROOF THE FOUNDATION RUNS TODAY:

Runs on a laptop: our existing extractor processed 12.5 min of real traffic (20 MB) in 82 s.

Packet-level parsing works: TLS/QUIC handshakes parsed; 308 of 342 TLS sites named (90.1%).

Tested: 62 automated tests in the codebase, 42 of them aimed at tampering and failure cases.

Cheap to run: CPU-only ONNX models on a ₹55,000 industrial PC. No GPU, no cloud.

**No world-model score is claimed yet. The table lists what we will measure and publish, win or lose.*



IMPACT & BENEFITS



SMART INDIA
HACKATHON
2026

NETSENTINEL VS. EXISTING TOOLS

Capability	Forecasts next stage	Learns time dynamics	ATT&CK stage mapping	Explains each output	Works offline
NetSentinel Foresight	Yes: K-step probability	Yes: world model	Yes, per forecast	Yes: SHAP + attention	Yes
Snort / Suricata	No	No: rules	Partial (rule tags)	Rule name only	Yes
ML flow classifiers (RF / XGBoost IDS)	No	No: one flow at a time	No	Sometimes	Yes
Commercial NDR / SIEM (e.g. Darktrace, Splunk)	Risk scores, not stage forecasts	Varies	Yes	Partial	Varies: often cloud

IMPACTS

The blind spot: SOC's protecting India's power, banking, telecom and government networks see an attack only after a stage completes, and CERT-In's 6-hour reporting clock starts after the damage is done.

What changes: a forecast minutes to hours ahead, naming the next stage and the host likely hit next.

For operators: an early warning while the attacker is still scanning.

For analysts: one forecast carrying the whole chain and its drivers, not scattered alerts.

For auditors: every forecast is signed and later checked against reality, so accuracy itself is auditable.

For the nation: runs inside the premises of NCIIPC-notified CII. No traffic data leaves the site.

Beyond this: hospitals, rail signalling, smart grids and IoT (tested on CIIoT2023).

USE CASES



Power & Energy Grid (CII)

Forecasts recon → lateral movement towards SCADA hosts before control is reached.



Government SOC & CERT-In

ATT&CK-mapped forecasts and signed evidence fit the 6-hour reporting rule.



Banking & Finance

Predicts brute-force → access → exfiltration and names the server at risk.



Enterprises, MSMEs & Colleges

CPU-only, offline, one SPAN port: early warning they can afford.



Security Benefits

Act before compromise: block, isolate or watch while the attacker is still in recon.

Knows the next target: response is focused on the host most likely hit next.

Catches the unseen: judges stage behaviour, not signatures.



Economic Benefits

Cheaper to stop early: containing at recon costs far less than cleaning up a breach.

Low hardware cost: one CPU box per segment, no GPU, no licence fees.

Less alert fatigue: one forecast per campaign, not dozens of alerts.



Technical Benefits

Offline & open source: code, weights and training configs included, reproducible.

Two levels of traffic: flow + packet features fused on a graph.

What-if simulation: try a defensive action in the model first.



RESEARCH & REFERENCES

CITATIONS

World Models

Ha & Schmidhuber, 2018

Learn an internal model, then roll it forward.

Latent Dynamics (RSSM)

Hafner et al., DreamerV3, 2023

The recurrent state-space model we adapt to network state.

Graph NIDS

Lo et al., E-GraphSAGE, IEEE NOMS 2022

Flows as graph edges; our encoder.

Attack Forecasting

Husák et al., IEEE COMST 2019

Survey of attack projection, prediction & forecasting.

Explainability

Lundberg & Lee, NeurIPS 2017 (SHAP)

Feature attribution for every forecast.

LINKS (Ctrl + Click on the blue words)



[Demo Video - Prototype](#)



[GitHub Repository](#)



[Documentation](#)



[Presentation](#)

NetSentinel Foresight · Standards & Knowledge Alignment

MITRE ATT&CK

Stage labels + technique IDs for every forecast (T1595, T1190, T1021, T1071, T1041...).

CAPEC

Attack-pattern chains → the stage-transition prior used by the dynamics model.

CVE / NVD

CVSS exposure of each host's open services raises its chance of being next.

NCIIPC · CII

Design fits NCIIPC guidance for protected systems: on-prem, passive, no data leaves.

CERT-In

ATT&CK-mapped, signed forecasts support the 6-hour incident reporting rule.

RFC 6962

Merkle transparency log (as used for web certificates) for tamper-evident forecast receipts.

MODEL SELECTION — PLANNED HEAD-TO-HEAD

Component	Compared against	Pass condition
World model · RSSM + Transformer	Logistic regression, XGBoost, plain LSTM, Markov chain	Beat LR on F1 at equal FPR, and lose accuracy when time is shuffled
Graph encoder · E-GraphSAGE	Flat feature vector	Better next-target accuracy from who-talks-to-whom
Rollout · 64 samples	Single one-step prediction	Calibrated probabilities (Brier / ECE)
Explainer	Attention only · SHAP · Integrated Gradients	Faithful: removing top drivers drops the risk
ATT&CK / CAPEC prior	No prior	Higher recall on held-out attack families

Same features and the same time-based split for every row. Results will be published whether our model wins or not.

DATASET IMPLEMENTATION

1. CIC-IDS2017/2018: flows + PCAP, day-wise attack timelines

2. CTU-13: 13 botnet scenarios, labelled NetFlow + PCAP

3. UNSW-NB15: 9 attack families, PCAP + flows

4. CICIOT2023: 33 IoT attacks (unseen-attack tests)

5. DARPA 2000 LLDOS 1.0/2.0: 5-phase multi-stage attack

6. LANL auth + red team: real lateral movement, 58 days

7. MITRE ATT&CK · CAPEC · CVE/NVD: knowledge prior

8. Own capture, IPv4 + IPv6: benign baseline (already collected)